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BLOCKCHAIN Based Voting SYSTEM

Second Year Project Synopsis Submitted by

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Project Overview

- A Blockchain Voting System is a decentralized and tamper-proof electronic voting solution.
- That leverages blockchain technology to ensure security, transparency, and integrity in elections.
- This system eliminates election fraud, enhances voter privacy, and provides verifiable results in real time.



About the Problem

➤ Problems Identified

- Election Fraud & Manipulation
- Lack of Transparency & Trust
- Security & Data Integrity Issues

➤ Issues or Problems

- Scalability Issues
- Voter Identity & Privacy
- Security Risks

➤ Need of Solution

- Enhanced Transparency & Trust
- Elimination of Election Fraud & Tampering
- Improved Security & Data Integrity

➤ Existing Solutions

- Uses **biometric authentication** (facial recognition, fingerprint scanning)
- Enables remote voting via a mobile app



Problem Statement

- By adopting blockchain in the distribution of database on e-voting systems can Reduce one of the cheating sources of database manipulation.
- Blockchain technology is one of solutions, because it embraces a distributed system and the entire database are owned by many users.
- Blockchain itself has been used in the e-voting, app voting system.
- Identity Verification Challenges: Ensuring one vote per person without compromising anonymity is complex.



Objectives

- To develop a secure and transparent voting system.
- To prevent election fraud and manipulation.
- To enable voters to verify their votes while maintaining anonymity.
- To ensure elections results are immutable and publicly auditable.



Methodology, Tools, and Techniques

➤ **Requirement Analysis**

Stake Holders

Government Elections Authorities

Voters

Secure voter registration and authentication

Immutable vote recording on the blockchain

➤ **Tools**

Cryptographic Security Tools

Smart Contract Development

Frontend & User Interface Development

➤ **Techniques**

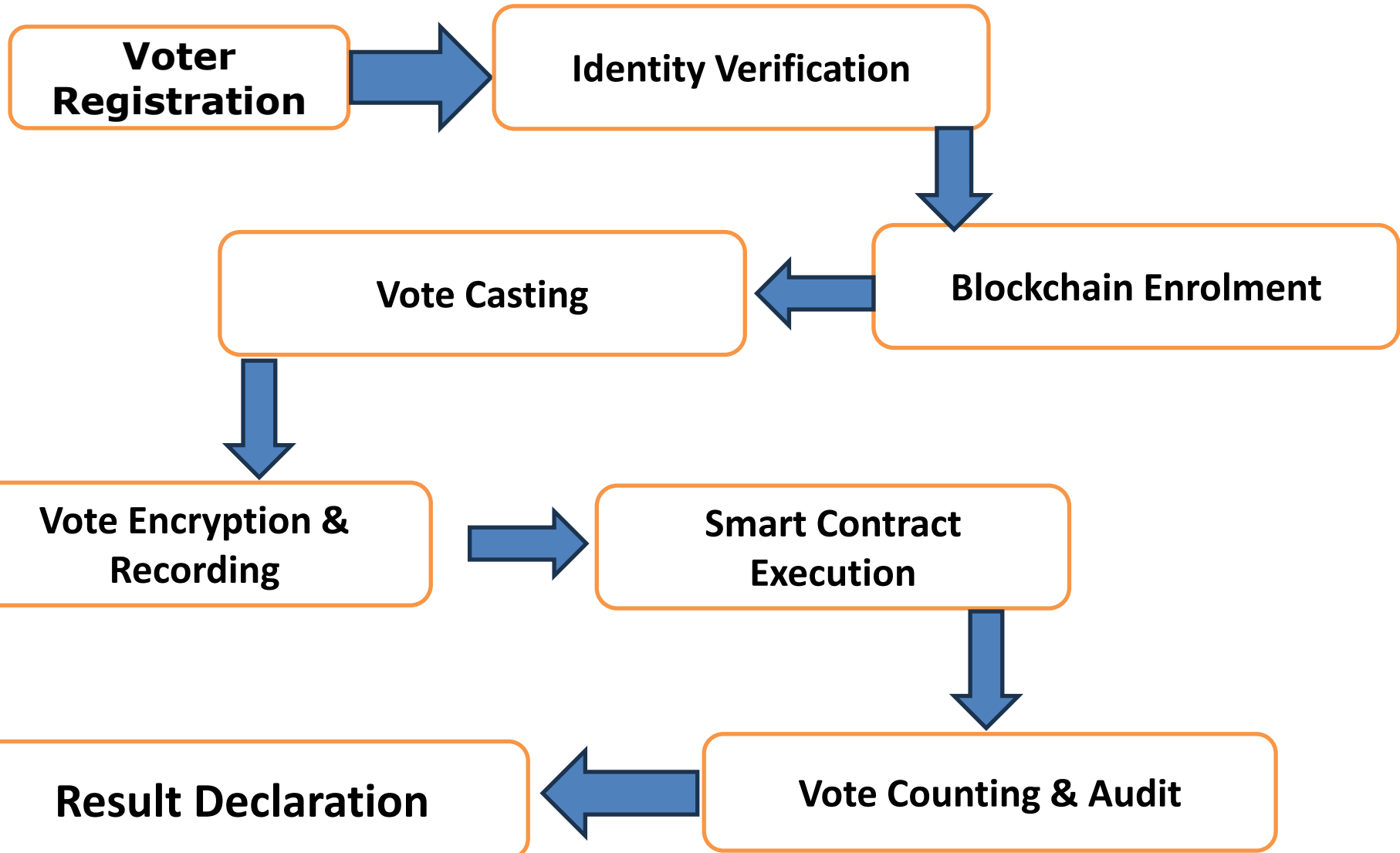
Secure Voter Authentication

Secure Voting & Anonymity Techniques

Performance Optimization Techniques



Methodology Flowchart



Project Timeline

- This 6-month timeline covers key stages of development.
- From requirement analysis to deployment and monitoring.
- It ensures a step-by-step approach for ensuring security, scalability, and user-friendliness of the blockchain-based voting system.



Expected Results & Impact

Expected Results:

The expected results from implementing a Blockchain-Based Voting System can be summarized in terms of security, transparency, efficiency, and user satisfaction.

- Enhanced Security
- Anti-Hacking Measures
- Transparency & Trust

Impact:

The Blockchain-Based Voting System has the potential to transform the way elections are conducted globally, creating profound effects on security, voter participation, efficiency, and government accountability

- Increased Security and Fraud Prevention
- End-to-End Encryption and Zero-Knowledge Proofs
- Protection Against Cyberattacks



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THANK YOU

